When Structure Hurts: Understanding Evidence Loss in Retrieval-Augmented Reasoning

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Abstract

Retrieval-augmented systems increasingly turn retrieved passages into sentence or claim lists, reasoning orders and relation graphs before generation. Such transformations can help, but they can also make evidence harder to read, spend context on formatting, or assert relations that the sources do not support. System-level comparisons cannot separate these effects, because retrieval, content and formatting change together.

We hold the evidence fixed. Gold supporting passages for multi-hop questions are presented in seven representations whose fact sets are verified identical per question; a second protocol fixes a fact ranking and a token budget. With Qwen2.5-7B-Instruct, we tested six contrasts, specified before data collection, on a disjoint sample of 4,976 MuSiQue and 2WikiMultiHopQA questions. Graph-derived ordering and explicit derived relation lines had no detectable effect on exact match; on 2WikiMultiHopQA both are equivalent to zero within ± 2 points. Splitting passages into sentence items lowered 2WikiMultiHopQA exact match by 1.7 points (95% CI $[-2.9, -0.5]$), almost entirely through more abstention.

Exploratory controls, added after this test, show where structure hurts. With order and line count held fixed, showing the `CONTRADICTS` relations predicted by an NLI classifier lowered exact match by 3.3 points on both datasets, mainly by making the reader abstain. The effect replicated on 1,000 fresh questions with real retrieval. Without the instruction that permits abstention, most representation differences shrank toward zero. For this reader, the measurable cost of structure came from asserted relations and from abstention behaviour, not from the reorganisation of facts.

1 Introduction

Retrieval-augmented generation (RAG) gives a language model access to text that its parameters do not contain, by retrieving passages and placing them in the prompt (Lewis et al., 2020). For multi-hop questions, as in MuSiQue, 2WikiMultiHopQA and HotpotQA (Trivedi et al., 2022; Ho et al., 2020; Yang et al., 2018), the passages are usually handed to the model as a list, and finding and linking the relevant sentences is left to the model.

A growing family of methods therefore transforms retrieved evidence before generation. Passages are decomposed into propositions or claims (Chen et al., 2024), organised into trees of summaries (Sarthi et al., 2024), or linked into graphs whose nodes, relations and paths are serialised into the prompt (Edge et al., 2024; Gutiérrez et al., 2024, 2025; Zhu et al., 2025; Guo et al., 2024; Chen et al., 2026). Each such transformation can also go wrong. Splitting a passage into sentences turns "It stars Anne Hathaway ..." into an item whose subject must be recovered from elsewhere; a token budget may cut the sentence that names the answer because the format spends tokens on identifiers and relation lines; and a relation predictor may assert a relation the sources never made. In our experiments an NLI classifier marked "The teaser of the film was released in February 2016" and "The trailer of the film was released in October 2016" as contradicting each other, and, shown that line, the model declared the evidence insufficient for a question it otherwise answered correctly.

These costs are hard to separate from the benefits in existing evaluations. When a graph-based system is compared with passage RAG, its retrieval, the content that reaches the reader and the way that content is formatted all differ at once (Xiang et al., 2026; Han et al., 2025; Zhou et al., 2025). The modular analysis of Nishida et al. (2026) shows that noisy extracted triples hurt GraphRAG. There too, however, extraction errors act through what is retrieved, so the effect of an in-

correct relation *as presented to the reader* remains unmeasured.

We argue that the effect of an evidence representation should be measured with the evidence held fixed, and that each component of a structured representation should be varied on its own: sentence segmentation, order, relation lines, and their provenance and labels. We therefore investigate the following question. When the same supporting facts are presented as passages, sentence items, ordered items or items with explicit relations, does answer accuracy change, and which component causes the change?

We present the gold evidence of each question in seven representations under two protocols. An *information-controlled* protocol verifies per question that every representation shows the same facts. A *budget-controlled* protocol fixes a representation-independent fact ranking and token budget, and records which facts each format can afford. After an exploratory pilot, we tested six contrasts, fixed before any generation, on a disjoint sample of 4,976 questions with Qwen2.5-7B-Instruct.

The confirmatory test was largely null for structure: graph-derived order and derived relation lines did not change exact match detectably. The one confirmed effect, a small cost of sentence items on 2WikiMultiHopQA, ran through abstention. We then added exploratory, fixed-order controls that change one factor at a time. They identify where structure hurts: showing the contradiction relations predicted by an NLI classifier costs about three exact-match points by making the reader abstain, while the order and the number of relation lines explain almost none of the effect. The pattern held on fresh questions with real retrieval. Removing the instruction that allows abstention removed most representation differences.

Our contributions are:

- a controlled framework that separates the facts a representation shows from its segmentation, order, relation lines, relation provenance and formatting overhead, with per-question verification of fact identity;
- a confirmatory test, specified before data collection, of order, relation lines and sentence segmentation on 4,976 held-out questions, with mostly null and equivalence results;
- exploratory fixed-order interventions showing that predicted contradiction relations harm a reader that is allowed to abstain, replicated with real retrieval and qualified by a prompt manipulation;
- a reproducible release of code, frozen question sets, run manifests and analysis outputs, and an audit mapping every number in this paper to its source.

2 Related Work

Passages, finer units and compression. Retrieval-augmented generation conditions a language model on retrieved text (Lewis et al., 2020), usually by concatenating passages in rank order. This presentation is not neutral: models use information in the middle of a long context less reliably than at its edges (Liu et al., 2024), and irrelevant passages change generation quality in ways that retrieval scores do not predict (Cuconasu et al., 2024). Dense X Retrieval indexes propositions and finds that finer units help retrieval and downstream QA under a fixed computation budget (Chen et al., 2024). Compression methods trade information for length explicitly (Jiang et al., 2024; Jeong et al., 2025); our budget protocol instead fixes the fact ranking, so that formatting cost is separated from content choice.

Graph-structured retrieval. GraphRAG builds an entity graph with community summaries (Edge et al., 2024), RAPTOR a tree of recursive summaries (Sarthi et al., 2024), and HippoRAG and HippoRAG 2 a knowledge graph traversed with personalised PageRank (Gutiérrez et al., 2024, 2025). KG^{2}RAG links chunks through fact-level relations (Zhu et al., 2025), LightRAG combines graph indexing with dual-level retrieval (Guo et al., 2024), and PathRAG retrieves, prunes and serialises relational paths (Chen et al., 2026). LinearRAG avoids relation extraction altogether, arguing that extracted relations are unreliable and costly (Zhuang et al., 2026). In all of these systems graph construction, retrieval and prompt formatting change together.

Analyses of when structure helps. The closest work to ours is the modular analysis of GraphRAG by Nishida et al. (2026). They vary triple extraction, community clustering and report generation on two templated factoid benchmarks with a GPT-4o reader, and find that the precision and coverage of extracted triples limit downstream accuracy,

that noisy LLM-extracted relations hurt both evidence and answer recall, and that simple template reports outperform LLM summaries. GraphRAG-Bench evaluates the whole pipeline on tasks of increasing complexity and identifies when graphs give measurable benefits (Xiang et al., 2026). A benchmark comparison of RAG and GraphRAG finds task-dependent strengths (Han et al., 2025), and a unified framework compares graph-based RAG methods under the same settings (Zhou et al., 2025). Dai et al. (2024) present the same knowledge-graph facts as linearized triples or as fluent text and find that the format matters and that model preferences differ.

These studies establish that structure does not help uniformly and that extraction errors propagate. In each of them, however, changing the structure also changes what the reader sees, because extraction, clustering and verbalisation determine the retrieved content, or, in the format comparison, because the units themselves change. Our study is narrower. We fix the sentences the reader sees, verify their identity per question, and vary one presentational factor at a time: segmentation into sentence items, order, explicit relation lines, the provenance of those relations (derived from annotations or predicted by a model), and their labels at a fixed number of lines. We also account for the tokens each format spends on itself under a budget, and we separate abstention from wrong answers. We do not measure whether structure improves retrieval, which these system-level studies address.

Datasets and attribution. We evaluate on MuSiQue (Trivedi et al., 2022) and 2WikiMultiHopQA (Ho et al., 2020), with HotpotQA (Yang et al., 2018), HoVer (Jiang et al., 2020) and FEVER (Thorne et al., 2018) as secondary pilot datasets. ALCE evaluates citations (Gao et al., 2023); we record citations but do not analyse them. Experiment 0 includes our reconstruction of the EG-RAG evidence-graph pipeline only as a retrieved-evidence baseline. Its unspecified components are our own choices (Appendix C), and no result of the original architecture paper is reused.

3 Problem Formulation

Structured pipelines change *which facts* reach the answerer and *how* they are arranged and annotated; we measure the two separately.

Facts and representations. For a question q with gold answers A_q we start from a fixed set of source passages; in our controlled experiments these are the dataset's gold supporting passages. Segmenting them into sentences gives the fact set $F_q = \{f_1, \ldots, f_n\}$, where each fact keeps the identifier of its source span. Because facts are sentences of the gold passages, not model-generated paraphrases, F_q contains no extraction error. A sentence is not an atomic proposition, however: some sentences need their passage to be read (*It stars* ...), so segmentation can make a fact harder to interpret without removing it. Model-based proposition extraction enters only through the retrieved-evidence systems of Experiment 0, and we never mix extracted claims into an information-controlled comparison. A representation ρ maps the facts, optionally with structure S_q derived from annotations or predicted by a model, to an evidence block

$$E_q^\rho = \mathrm{render}_\rho\big(\pi_\rho(F_q),\, S_q\big), \qquad (1)$$

where π_ρ orders and groups the facts and render_ρ serialises them, possibly with a list of relations. We write $\mathrm{facts}(E)$ for the facts that appear in a block, and $\mathrm{added}(\rho)$ for the information that the block carries beyond them. We distinguish three kinds of added information that structured pipelines usually change together: the *order* π_ρ of the items; explicit *relation lines* $L_\rho \subseteq F_q \times F_q \times \mathcal{T}$, each asserting a typed relation (dependency, support, contradiction, duplication, supersession) between two items; and the *provenance* of those relations, derived deterministically from dataset annotations or predicted by a model. Our controls change one of these at a time, and include a neutral label that asserts no relation type while keeping the same pairs and number of lines. A fixed answerer M with one prompt template τ, identical for every representation, produces $\hat{a} = M(\tau(q, E_q^\rho))$ with greedy decoding. The template allows the answer *insufficient evidence*, so each outcome is one of three states: correct (C), answered but wrong (W), or not answered (A, an abstention or an unparseable reply). Exact match counts A as incorrect; we report the transitions between states separately, because a representation can change accuracy by changing the willingness to answer.

Protocol A: information-controlled. Protocol A compares representations whose fact sets are identical, $\mathrm{facts}(E_q^\rho) = F_q$ for every ρ, which we

verify per question by comparing identifiers and text digests. An accuracy difference then cannot come from a missing fact, only from $\mathrm{added}(\rho)$ or formatting tokens. We record the content tokens $c_q = \sum_j |\mathrm{text}_j|$ and the overhead $o_q^\rho = |E_q^\rho| - c_q$ separately.

Protocol B: budget-controlled. Under a budget $B_q = \beta\, c_q$ we rank the facts once, with BM25 against the question, independently of the representation and without gold labels, and each condition keeps the longest prefix of the ranking whose rendering fits,

$$k_\rho = \max\{k : \ |\mathrm{render}_\rho(\pi_\rho(F_q^{\le k}), S_q)| \le B_q\}. \quad (2)$$

Under the *per-representation* policy each representation keeps its own k_ρ, so verbose formats keep fewer facts; under the *shared* policy all keep $\min_\rho k_\rho$ and show the same facts. Facts are never cut mid-sentence. For every dropped fact we record whether it is a gold supporting fact and whether it contains the answer string; this is our measure of information loss.

Objective. For a contrast between ρ and ρ' we estimate the paired difference in question-level exact match over the questions Q available in both conditions, with unanswered questions scored as incorrect:

$$\Delta_{\rho'-\rho} = \tfrac{1}{|Q|} \textstyle\sum_{q\in Q} \big(\mathrm{EM}(q,\rho') - \mathrm{EM}(q,\rho)\big). \quad (3)$$

4 Experimental Framework

Figure 1 (Appendix A) shows the pipeline in the order in which a question passes through it.

Questions and structure. Questions come from frozen samples of the development splits, drawn with a fixed seed before any generation and stored with the hashes of the dataset files; the pipeline refuses to run on any question outside a frozen file or on the questions used during prompt development. The MuSiQue decomposition and the 2WikiMultiHopQA evidence triples give an order of reasoning steps, from which we *derive* dependencies between facts by rule. Where a dependency can be anchored to a fact by an exact string match we call it *exact-anchor derived*, otherwise *fallback-anchor derived*. No edge in this study is human-verified.

ID	Rendering of the same fact set
R0	one item per gold passage, dataset order
R1	one item per sentence, dataset order
R2	R1 shuffled (seeds 0, 1, 2; averaged per question)
R3	R1 grouped under `Step` k`:` in reasoning order
R5	R1 in graph-derived (topological) order
R4	R5 plus a `Relations:` block of derived dependencies
R4P	R4 with model-predicted relations and order
Fixed-order controls (exploratory)	
R4P_FO	R5 order + predicted relation lines
R4P_FO_NC	R4P_FO without `CONTRADICTS` lines
R4N_FO	R4P_FO pairs, neutral label `RELATED_TO`
R5P	predicted order, no relation lines
R4P_NC	R4P without `CONTRADICTS` lines

Table 1: Representations. Every item carries a per-prompt identifier (`[E1]`, ...) and its source title.

Retrieved-evidence baselines (Experiment 0, pilot). Five systems answer from the top four passages of each question's own pool: BM25 (B0), reranked (B1), extracted claims of the reranked passages (B2), a predicted reasoning path over them (B3), and our reconstruction of the EG-RAG evidence-graph pipeline (B4; Appendix C). Their evidence budgets differ, so we report them as systems.

Representations (Experiments 1 and 2). Table 1 defines the representations; Appendix B shows examples. R5 orders the sentence items by a topological order of the derived dependency graph at passage level, and R4 adds a `Relations:` block listing the derived dependencies. R5−R1 therefore measures the effect of the graph-derived order without relation lines, and R4−R5 the effect of explicit relations at a fixed order. R4P renders the same facts with *predicted* relations: typed relations (support, contradiction, duplication, supersession) from an NLI cross-encoder over candidate pairs, plus a title-mention dependency heuristic, with the order recomputed from the predicted graph.

Selection, generation and evaluation. Under Protocol A every representation shows the complete fact set; under Protocol B the budget is $\beta \in \{1.0, 0.5\}$ times the content tokens. A budget of $1.0\times$ the content tokens is already smaller than the full rendering of any representation that spends tokens on identifiers, titles or relation lines. The prompt (Appendix A) is identical for every representation, asks for an answer and the iden-

tifiers of the items used, and allows the answer *insufficient evidence*. A guard checks for every record that the prompt equals the template filled with the evidence block and the question, so no gold answer can enter the prompt through another channel. Exact match (EM) after SQuAD normalisation (Rajpurkar et al., 2016), maximised over aliases, is the primary metric for QA, and verdict accuracy for HoVer and FEVER. Secondary metrics are token F1, coverage (the fraction of questions with a parsed, non-abstaining answer), answer-in-evidence, and evidence and overhead tokens. Every attempted question stays in the denominator.

Fixed-order relation controls (added after the confirmatory test). R4P differs from R4 in three ways at once: its relation lines are predicted rather than derived, there are more of them, and its item order is recomputed from the predicted graph. To separate these factors we built five further representations of the same facts (Table 1, lower part). R4P_FO shows the predicted relation lines at the derived order of R5 and R4; R4P_FO_NC removes only the predicted `CONTRADICTS` lines from it; R4N_FO keeps exactly the predicted pairs and number of lines but replaces every label with the neutral `RELATED_TO`; R5P uses the predicted order without any lines; and R4P_NC is R4P without its contradiction lines. Each pairwise contrast between these conditions changes one factor on identical facts and an identical prompt template. Because they were designed after the confirmatory result and run on the confirmatory questions, we treat them as exploratory interventions.

Retrieved evidence (added after the confirmatory test). To test whether the presentation effects persist when retrieval is imperfect, we drew a fresh sample of 500 MuSiQue and 500 2WikiMultiHopQA questions, disjoint from all earlier samples. For each question a shared, gold-free retrieval stage (BM25 over the question's pool, cross-encoder reranking of the top ten, top four kept) fixes the passages before any representation is built, and every representation renders exactly their sentences. R4 and R5 use the annotation-derived edges whose two endpoints were retrieved, so they test correct structure on incomplete evidence; R4P, R4P_NC and R4P_FO use predicted relations. Gold labels are used only afterwards, to stratify questions by whether retrieval returned all, some or none of the supporting sentences.

Prompt factor (added after the confirmatory test). Because the answer instruction offers *insufficient evidence*, abstention may mediate representation effects. On a fixed random subset of 1,000 confirmatory 2WikiMultiHopQA questions we repeated R0, R1, R4 and R4P with the same prompt minus that sentence, and analyse the prompt as a separate factor.

Perturbations (Experiment 2, pilot). Interventions on the R1 and R4 renderings remove required facts, retype relations, permute items, insert non-gold distractor sentences, strip titles, or corrupt a fact (Appendix D). Experiment 3 (component attribution) requires human-verified structure and was not run, and we did not evaluate an adaptive representation selector.

5 Experimental Setup

Questions. The exploratory pilot (seed 1, fixed before any generation) has 100 MuSiQue and 100 2WikiMultiHopQA questions, plus 50 each from HotpotQA, HoVer and FEVER; the fifteen questions used in prompt development are excluded everywhere. After the pilot we wrote a confirmatory protocol and committed it to version control, and only then drew a *confirmatory sample* with a different seed (2) from the questions that the pilot and prompt development had not used: 1,744 MuSiQue questions (hop 2/3/4: 687/686/371, the last stratum exhausting the eligible pool) and 3,232 2WikiMultiHopQA questions (808 per type). Its size is the number of questions needed for 80% power to detect a difference of 0.03 exact match at $\alpha = 0.05/6$, computed from the pilot standard deviation of paired differences inflated by 20%, maximised over the three contrasts of each dataset. The confirmatory run uses Protocol A only. The protocols were committed to our repository with timestamps before the samples they govern were drawn; they were not lodged with an external registry, so we call them *pre-specified*. The analyses added after the confirmatory result (fixed-order controls on the confirmatory questions, the prompt factor on a fixed random subset of 1,000 confirmatory 2WikiMultiHopQA questions, and the retrieved-evidence experiment on a fresh sample of 500 MuSiQue and 500 2WikiMultiHopQA questions drawn with seed 3 from the remaining pool) were themselves specified in a committed protocol before they were run, but we treat all of them as ex-

ploratory.

Answerer. The answerer is Qwen2.5-7B-Instruct (Qwen Team, 2024) at the pinned revision `a09a3545`, in bfloat16, with greedy decoding, at most 64 new tokens and at most 4,096 input tokens. Auxiliary models are bge-small-en-v1.5 (dense retrieval in B4), ms-marco-MiniLM-L6-v2 (reranking), nli-deberta-v3-xsmall (relation prediction) and nli-deberta-v3-small (grounding), all pinned (He et al., 2023). Experiment 0 also uses the 7B model as its claim extractor.

Hardware and batch size. All generations run on a single NVIDIA RTX 5000 Ada GPU (32 GB; CUDA 13.0, PyTorch 2.12.1, Transformers 5.12.1) with batch size 1; all runs together used about 12 GPU-hours. In a preflight on questions excluded from all reported runs, batched bfloat16 greedy decoding changed 3 of 8 outputs of the 7B model, including one answer, whereas unbatched outputs were identical across runs.

Statistics. The unit of analysis is the question. For each contrast we report the paired difference in question-level EM over the questions available in both conditions, with a 95% percentile bootstrap interval (2,000 resamples, seed 0) and a two-sided bootstrap p-value (Efron and Tibshirani, 1993); R2 seeds are averaged within a question before pairing. Every contrast also reports the number of questions whose two evidence blocks are byte-identical (for example, R5 equals R1 when the graph-derived order coincides with the dataset order), the same estimate restricted to the remaining *informative* pairs, and the paired difference in coverage. Protocols, datasets, budgets, policies and edge strata are never pooled. All these procedures, the primary metric and the primary contrasts were fixed in a pilot protocol before any generation. The confirmatory family consists of the six primary tests (three contrasts on two datasets) on the confirmatory sample, with Holm correction (Holm, 1979) at a family-wise α of 0.05; the protocol assesses equivalence to zero by whether the 90% bootstrap interval lies inside a margin of ± 0.02 fixed before the confirmatory sample (a per-test TOST). In this revision we additionally apply a paired TOST with Holm correction over the six tests; the two procedures agree for every contrast. Everything else, including all pilot results, the secondary contrasts on the confirmatory sample, and the analyses that we added after seeing the confirmatory result, is exploratory and reported as such. An interval that contains zero is reported as inconclusive, never as evidence of equivalence.

6 Results

We first report the confirmatory test, the only part of the study designed to support a claim, and then the exploratory analyses added after it. Intervals are paired 95% bootstrap intervals over questions, and every difference is stated as the first-named condition minus the second.

6.1 Confirmatory test

Table 2 reports the six pre-specified tests on 1,744 MuSiQue and 3,232 2WikiMultiHopQA questions. All 44,784 records were generated without errors, and the fact sets were verified identical across representations for every question.

Sentence items. One test rejects its null hypothesis after Holm correction. On 2WikiMultiHopQA, presenting the gold passages as sentence items (R1) rather than passages (R0) lowers exact match by 0.017 (95% CI $[-0.029, -0.005]$, $p_{\text{Holm}} = 0.018$). On MuSiQue the same contrast is -0.001 $[-0.014, +0.014]$ and equivalent to zero within ± 0.02.

Order and relation lines. Neither structural contrast is significant on either dataset. The graph-derived order (R5−R1) gives -0.011 and -0.008, and the derived relation lines at a fixed order (R4−R5) give $+0.010$ and $+0.006$ (MuSiQue, 2WikiMultiHopQA). On 2WikiMultiHopQA both are equivalent to zero within ± 0.02; on MuSiQue they are inconclusive.

Caveats. Many pairs are byte-identical (R5 = R1 for 1,858 of 3,232 2WikiMultiHopQA questions); on the remaining pairs the structural estimates stay within ± 0.02. All structure here is derived from annotations, not human-verified.

6.2 Abstention, not accuracy, carries the confirmed effect

Each question's outcome is correct (C), answered but wrong (W), or not answered (A). Table 12 (Appendix G) decomposes each paired difference into transitions between these states. On 2WikiMultiHopQA the sentence format raises the abstention rate by 5.2 points $[4.1, 6.3]$, while accuracy among questions answered under both formats is unchanged ($+0.002$, $[-0.013, +0.017]$). The new

Data	Contrast	n	ΔEM	95% CI	p	p_{Holm}	+/−	Ident.	Inform. Δ	ΔCov.	Equiv.
MuSiQue	R1−R0	1744	-0.001	[-0.014, +0.014]	0.974	0.974	70/71	3	-0.001	+0.009	yes
MuSiQue	R5−R1	1744	-0.011	[-0.026, +0.003]	0.122	0.535	68/88	522	-0.016	+0.003	no
MuSiQue	R4−R5	1744	+0.010	[-0.003, +0.024]	0.117	0.535	75/57	0	+0.010	+0.004	no
2Wiki	R1−R0	3232	-0.017	[-0.029, -0.005]	0.003	0.018	185/239	44	-0.017	-0.052	no
2Wiki	R5−R1	3232	-0.008	[-0.018, +0.002]	0.107	0.535	126/152	1858	-0.019	-0.012	yes
2Wiki	R4−R5	3232	+0.006	[-0.002, +0.015]	0.174	0.535	103/84	879	+0.008	-0.011	yes

Table 2: Confirmatory test of the six pre-specified primary contrasts (Protocol A, EM, disjoint sample; `docs/CONFIRMATORY_PROTOCOL.md`). p: two-sided paired bootstrap; p_{Holm}: Holm-adjusted over the six tests. Equiv.: 90% CI inside ± 0.02 (pre-specified secondary check). Source: `F_exp1_contrasts.csv`.

abstentions fall on previously correct answers (87) and, more often, on previously wrong ones (163), and the net is the confirmed -0.017.

The derived relation lines show a mirror image on 2WikiMultiHopQA. They raise accuracy among answered questions ($+0.014$, $[+0.005, +0.024]$) but also raise abstention by 1.1 points, and the two offset each other in exact match. Abstention is common in general: with complete gold evidence the model leaves 47% of MuSiQue and 19% of 2WikiMultiHopQA questions unanswered under R0.

6.3 Fixed-order controls: contradiction lines, not order (exploratory)

The secondary contrast R4P−R4 on the confirmatory sample, predicted relations instead of derived ones, costs 3.6 (2WikiMultiHopQA) and 4.5 (MuSiQue) points. Because R4P also changes the order and the number of lines, we ran the fixed-order controls of §4 on the same questions (Table 3; Figure 4 in Appendix G).

The relation labels carry the effect. With the order fixed, showing the predicted `CONTRADICTS` lines rather than removing them costs 3.3 points on both datasets and raises abstention by 6 points. Replacing every predicted label by a neutral `RELATED_TO` at the same pairs and the same number of lines recovers a similar amount (3.7 and 3.0 points).

Order and line count do not. The predicted order itself changes nothing on 2WikiMultiHopQA and helps slightly on MuSiQue ($+0.018$). Neutral lines alone cost nothing on 2WikiMultiHopQA and 1.5 points on MuSiQue.

The effect is specific to contradiction lines. In questions whose predicted block contains no contradiction line, relabelling the other predicted relations to neutral changes exact match by -0.003 on both datasets. Among questions with one or two contradiction lines, showing them costs 5–7 points (Appendix G). Contradiction lines turned 89 correct 2WikiMultiHopQA answers into abstentions and only 9 the other way (MuSiQue: 52 and 2). These contrasts are interventions on identical facts, but they were designed after the confirmatory result and run on its questions.

6.4 Retrieved evidence (exploratory, fresh questions)

Table 4 repeats the contrasts on 1,000 fresh questions, where every representation renders the same four reranked passages.

The relation effects replicate. Predicted relations lower exact match by 3.6 (MuSiQue) and 3.0 (2WikiMultiHopQA) points, and contradiction lines by 2.2 and 2.4 points, with 4–7 points more abstention. Both effects are largest when retrieval returned all supporting sentences: R4P−R4 is -0.094 on MuSiQue and -0.042 on 2WikiMultiHopQA in that stratum. When nothing relevant was retrieved, no representation matters.

Sentence items are inconclusive. R1−R0 is -0.026 $[-0.054, +0.002]$ on 2WikiMultiHopQA, the direction of the confirmed effect, and $+0.014$ on MuSiQue.

Derived structure rarely survives retrieval. An annotation edge needs both endpoint sentences retrieved, so R5 equals R1 for 498 of 500 2WikiMultiHopQA questions. Correct structure on incomplete evidence therefore remains untested here.

6.5 The prompt factor (exploratory)

Table 14 (Appendix G) repeats R0, R1, R4 and R4P on 1,000 confirmatory 2WikiMultiHopQA questions without the sentence that offers *insufficient evidence*. Abstention falls from 20–33% to under 1%, and exact match

Factor changed (first − second)	2Wiki ΔEM	95% CI	ΔAbst.	Ident.	MuSiQue ΔEM	95% CI	ΔAbst.	Ident.
contradiction lines (on vs. off), derived order	-0.033	[-0.042, -0.025]	+0.061	1198	-0.033	[-0.043, -0.023]	+0.062	734
typed vs. neutral labels, same pairs and line count	-0.037	[-0.045, -0.029]	+0.056	530	-0.030	[-0.040, -0.020]	+0.048	410
neutral lines vs. none (count/overhead only)	+0.007	[-0.003, +0.017]	+0.019	530	-0.015	[-0.026, -0.004]	+0.038	410
predicted vs. derived order, no lines	+0.002	[-0.001, +0.005]	-0.003	3057	+0.018	[+0.006, +0.031]	+0.003	704
predicted vs. derived order, predicted lines	+0.000	[-0.003, +0.003]	-0.003	3057	+0.010	[-0.002, +0.023]	-0.001	704
original R4P−R4 (order and lines)	-0.036	[-0.045, -0.027]	+0.061	818	-0.045	[-0.060, -0.029]	+0.089	94

Table 3: Fixed-order relation controls on the confirmatory questions (exploratory interventions designed after the confirmatory result; n=3,232 2Wiki, 1,744 MuSiQue). Each row changes one factor on identical facts. ΔAbst.: paired change in the abstention rate; Ident.: byte-identical prompt pairs. Source: `revision/controls/relation_control_effects.csv`.

Data	Contrast	Ident.	ΔEM	95% CI	ΔAbst.
MuSiQue	R1−R0	0	+0.014	[+0.000, +0.028]	-0.006
MuSiQue	R5−R1	451	+0.010	[+0.000, +0.020]	-0.008
MuSiQue	R4−R5	342	+0.002	[-0.012, +0.016]	-0.002
MuSiQue	R4P−R4	72	-0.036	[-0.056, -0.020]	+0.068
MuSiQue	contradiction lines on−off	131	-0.022	[-0.038, -0.008]	+0.040
MuSiQue	R4P_FO−R5	86	-0.022	[-0.038, -0.008]	+0.052
2Wiki	R1−R0	5	-0.026	[-0.054, +0.002]	+0.072
2Wiki	R5−R1	498	+0.000	[+0.000, +0.000]	+0.000
2Wiki	R4−R5	322	+0.010	[-0.008, +0.028]	-0.006
2Wiki	R4P−R4	84	-0.030	[-0.054, -0.008]	+0.066
2Wiki	contradiction lines on−off	110	-0.024	[-0.046, -0.004]	+0.068
2Wiki	R4P_FO−R5	66	-0.016	[-0.042, +0.010]	+0.052

Table 4: Retrieved evidence, Protocol A (exploratory; fresh disjoint sample, n=500 per dataset; all representations render the same four reranked passages). Ident.: byte-identical prompt pairs. Full table with Protocol B in Appendix G. Source: `revision/retrieval/retrieval_contrasts.csv`.

rises by 8–13 points for every representation. Within this prompt, the sentence-item contrast is -0.004 $[-0.026, +0.018]$, and predicted relations cost -0.017 $[-0.035, 0.000]$, now through wrong answers rather than abstentions. The prompt×representation interactions ($+0.019$ for both contrasts) have intervals that touch or include zero. The representation effects of the main study are therefore conditional on an instruction that lets the model decline, and most of them act through that option.

7 Discussion

Reorganising the same facts changed little. With the facts held fixed, neither the graph-derived order nor explicit derived relation lines changed exact match by more than about one point. The structure we tested is derived from annotations and therefore mostly correct, and it accompanies facts that already suffice. For a 7B instruction-tuned reader given the supporting facts, the benefit attributed to graph-structured evidence in system-level comparisons is unlikely to come from relation lines or ordering as such.

When structure hurts: asserted contradictions. The clearest harm came from relation lines that assert something the sources do not. Gold supporting passages are, by construction, consistent evidence for one answer. Yet the NLI classifier marks at least one pair of them as contradictory in 63% of 2WikiMultiHopQA prompts, for example the teaser and trailer dates of one film. An automatic triage suggests that most such pairs concern different entities (40%) or different dates (20%) and are therefore likely false. That is an inference, and whether the harm requires the contradictions to be *false* awaits the human labels we have prepared. What the controls establish is that *showing* these lines makes the reader abstain. Complementing Nishida et al. (2026), who find that noisy extracted relations hurt GraphRAG through retrieval, such relations also hurt when merely displayed next to fixed, correct evidence.

Abstention is the channel. Every clear effect in this study ran mostly through the model's willingness to answer, and without the instruction that permits abstention the representation differences shrank toward zero. Some formats make an instruction-tuned reader more cautious, because the evidence looks less coherent or is said to conflict. Accuracy alone conflates this caution with reasoning failures, and the transition analysis separates them.

8 Conclusion

Holding the evidence fixed, we found no detectable effect of graph-derived order or derived relation lines for Qwen2.5-7B-Instruct on 4,976 held-out questions, and a small, abstention-driven cost of sentence items on 2WikiMultiHopQA. Exploratory fixed-order controls located where structure hurts: predicted contradiction relations cost about three exact-match points by making the reader abstain, an effect that replicated with real retrieval and largely vanished when abstention was not offered.

Limitations

One reader, one prompt. All generations come from Qwen2.5-7B-Instruct with greedy decoding. The confirmed effects act mainly through abstention under a prompt that offers *insufficient evidence*, and they largely vanish without that option. Other model families, larger models, or other instructions may behave differently. The confirmatory sample is disjoint from the pilot, but it is not an independent replication with a different reader, dataset or team. A second-reader replication is prepared but was not run.

Derived, not verified, structure. Every dependency edge is derived from dataset annotations by rule, and for a substantial fraction of questions the claim-level endpoint is a fallback guess. No relation in this study is human-verified. We did not run the component-attribution experiment, which requires verified structure, and we report no relation precision or recall. The predicted contradictions are judged false only by automatic triage. Two blinded annotation packages are prepared but not completed.

Sentence units, not propositions. Our fact units are dataset sentences. Some depend on their passage to be interpreted, and they are not LLM-extracted atomic propositions. Our results concern sentence-item presentation, not the end-to-end effect of claim extraction, which we observed only at pilot scale with unequal budgets.

Gold evidence and retrieval. The confirmatory test and the controls use gold supporting passages. The retrieved-evidence replication uses 500 questions per dataset; its MuSiQue sample is almost entirely 2-hop, because earlier samples exhausted the longer questions. Annotation-derived structure rarely survives top-4 retrieval, so correct structure on incomplete evidence remains untested.

Exploratory analyses. The fixed-order controls, the prompt factor and the retrieved-evidence experiment were designed after the confirmatory result. They were specified before they were run, but they are not multiplicity-corrected and are not confirmations. The controls run on the confirmatory questions. A planned regression of the contradiction effect on line counts was abandoned because its predictors are collinear.

Metric and pilot scale. Exact match is strict; a diacritic-folding check changed few records and no conclusion. Protocol B, Experiment 0 and the perturbations were run only on the 100-question-per-dataset pilot. The pilot overestimated the one effect we later confirmed by a factor of four.

Compute. All runs used one 32 GB GPU with batch size 1, about 12 GPU-hours in total.

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A Pipeline and prompts

Every Experiment 0–2 answer uses the same prompt (version `qa-v2`); only the evidence block differs between representations. The system message is *You answer using only the evidence you are given.* The user message is:

```
Use only the evidence below to answer the question.
Each evidence item has an ID such as [E1]. Items may
be grouped into steps or followed by a list of
relations between items.

Evidence:
{evidence}

Question: {question}

Reply in exactly two lines:
Answer: <short answer: a name, date, number, short
phrase, or yes/no>
Evidence: <IDs of the items you used, e.g. E1, E2>
If the evidence is not sufficient, reply
"Answer: insufficient evidence".
```

The assistant turn is prefilled with `Answer:`. HoVer and FEVER use the same template with *Claim* in place of *Question*, ask for one of the dataset's labels, and have no abstention line. The claim extractor of Experiment 0 (version `extract-v2`) uses the system message *You split sentences into short, simple facts.* and asks for one fact per line, naming the subject instead of using pronouns and using only information stated in the sentence. Its output is prefilled with `- `.

B Serialisation examples

The following renderings are produced by the experiment code for a MuSiQue development question that was used only during prompt development and is excluded from every reported run: *What character in Princess Diaries does the actress play who was also a cast member in On Golden Pond?* (gold: *Queen dowager Clarisse Renaldi*). Long sentences are abbreviated here with "…".

R0 (passages).

```
[E1] (source: The Princess Diaries
(film)) The Princess Diaries is a 2001
American teen romantic comedy ... It
stars Anne Hathaway ... ruled by her
grandmother Queen dowager Clarisse
Renaldi (Julie Andrews). ...
[E2] (source: On Golden Pond (2001
film)) On Golden Pond is a 2001
American television adaptation of the
play starring Julie Andrews and
Christopher Plummer. ...
```

R1 (sentences, dataset order).

```
[E1] (source: The Princess Diaries
(film)) The Princess Diaries is a 2001
American teen romantic comedy ...
[E2] (source: The Princess Diaries
(film)) It stars Anne Hathaway ...
Queen dowager Clarisse Renaldi (Julie
Andrews).
[E3] (source: The Princess Diaries
(film)) The film also stars ...
[E4] (source: On Golden Pond (2001
film)) On Golden Pond is a 2001
American television adaptation of the
play starring Julie Andrews and
Christopher Plummer.
[E5] ... [E6] ...
```

Item E2 begins with a pronoun. Separated from its passage, its subject is recoverable only through the source title that every item carries. We record such context-dependent facts per question; they are one way in which an information-equivalent transformation can still change what is easy to read.

R3 (steps), R5 (graph order) and R4 (graph order plus relations). R3 groups the same six items under `Step 1:` (the three *On Golden Pond* sentences) and `Step 2:` (the three *Princess Diaries* sentences), following the MuSiQue decomposition. R5 lists the same items in the same order without step headers. R4 is R5 followed by

```
Relations:
E5 DEPENDS_ON E1
```

i.e. the sentence naming the character depends on the sentence establishing that Julie Andrews was in *On Golden Pond*. This edge is derived from the MuSiQue decomposition with an exact anchor; it is not human-verified.

C Reconstruction of the EG-RAG baseline

B4 follows the stage order of the EG-RAG algorithm: hybrid retrieval, reranking, claim extraction, relation prediction, belief and utility propagation, conflict resolution, budgeted subgraph selection and serialisation. Every component that the original description leaves unspecified is our choice:

- BM25 plus bge-small-en-v1.5 with min–max normalisation and equal weights;
- ms-marco-MiniLM-L6-v2 reranking;
- the 7B generator as claim extractor, with a lexical-support filter;

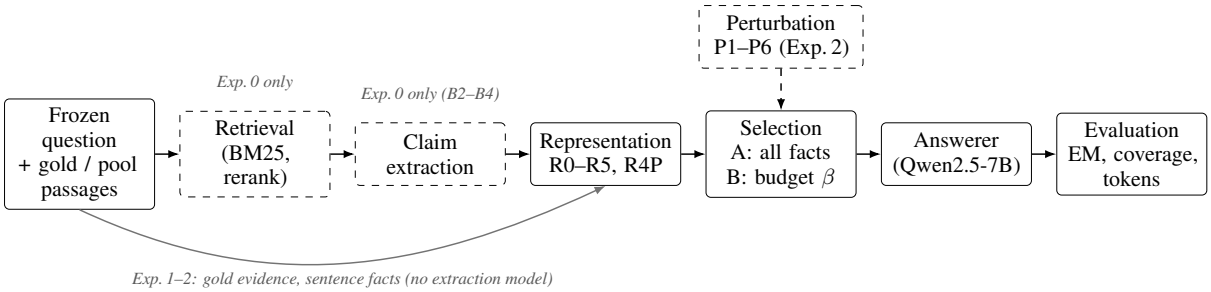


Figure 1: The evidence pipeline. Experiment 1 starts from the gold passages and varies only the representation (Protocol A), or the representation under a fixed fact ranking and token budget (Protocol B). Experiment 0 runs the retrieved-evidence pipeline for five systems; Experiment 2 corrupts the evidence of two representations. Dashed boxes are used only by some conditions.

- nli-deberta-v3-xsmall for support, contradiction and duplication, with thresholds 0.7 and 0.9;
- a title-mention heuristic for dependencies;
- years as the only temporal signal;
- our own feature weights, a 450-token budget and at most 12 claims;
- greedy connected selection.

The full ledger is in the repository (`docs/NOVELTY_AND_BASELINES.md`). B4 is therefore a reconstruction under our choices. Its results say nothing about the original implementation, and no result of the original paper is reused.

D Perturbations

Experiment 2 applies each intervention to the R1 and R4 renderings of the gold evidence with a recorded seed (0). The rates are $\{0.2, 0.4\}$ for P1–P4 and 0.4 for P5 and P6, each against the same question's rate-0 record, and $k = \lceil \text{rate} \cdot n_{\text{eligible}} \rceil$ items are affected.

- **P1** removes required (gold-support) facts.
- **P2** retypes or flips derived relations (R4 only).
- **P3** permutes item positions.
- **P4** inserts real non-gold sentences from the question's own candidate pool, chosen by BM25 similarity (*hard*), and never sentences containing a gold answer.
- **P5** strips source titles.
- **P6** deletes a negation, or replaces an entity in a fact with *it*.

Every changed item is recorded before and after. P1 and P6 can remove the answer and are therefore not label-preserving; we report their answer-change and abstention rates but no accuracy change. An intervention is unavailable when nothing is eligible:

- P2 on R1 or on questions without edges;
- P3 with a single item;
- P4 on HoVer and FEVER, which have no candidate pool;
- P6 when no fact contains a negation;
- P7 everywhere, since no verified temporal data exist.

These cases are counted and never scored as zero.

E Details of the exploratory pilot results

Evidence loss under a token budget (pilot). Protocol B was run only in the exploratory pilot (Figure 3 and Table 9, Appendix F). Its accounting of what each condition showed is deterministic and does not depend on the generator. Under a budget equal to the content tokens of the fact set, with each representation keeping as much of the shared ranking as it can afford, R0 keeps 0.83 (2WikiMultiHopQA) and 0.81 (MuSiQue) of the facts, whereas R1 keeps 0.64 and 0.65 and R4 keeps 0.63 and 0.62, because their identifiers, titles and relation lines cost 104–108 (2Wiki) and 81–88 (MuSiQue) tokens of overhead against 28 and 32 for R0. The verbose formats also lose the facts that matter: R0 drops 0.11 answer-bearing

facts per question on 2WikiMultiHopQA against 0.33 for R1 and 0.34 for R4 (MuSiQue: 0.19, 0.28 and 0.35), so that the answer string remains in 0.85 of R0 blocks but only 0.66 of R1 and 0.64 of R4 blocks. The corresponding accuracy differences are smaller than these evidence losses and inconclusive at $n = 100$. For example, R1−R0 is -0.05 $[-0.11, +0.01]$ on 2WikiMultiHopQA, and R4−R5 is -0.05 $[-0.11, +0.01]$ on MuSiQue. Under the shared policy, every representation shows the same facts and pays the same evidence loss. Halving the budget reduces exact match for every representation to 0.08–0.19.

Retrieved-evidence baselines (pilot). Table 10 (Appendix F) reports Experiment 0 on the 200 primary pilot questions. Reranking (B1) improves on BM25 (B0) on MuSiQue by $+0.06$ $[0.00, +0.12]$, and the 2WikiMultiHopQA difference is inconclusive. Extracted claims (B2) and predicted reasoning paths (B3) are indistinguishable from reranked passages in exact match; the extracted claims keep the answer string as often as the passages they come from (0.78 and 0.44), but at roughly three times the evidence tokens (1,390–1,711 against 501–538), because every passage sentence becomes several facts. Our reconstruction of EG-RAG (B4) uses a similar number of tokens to B1 (442–489 against 501–538) but is less accurate, by -0.15 $[-0.23, -0.07]$ on 2WikiMultiHopQA and -0.09 $[-0.15, -0.04]$ on MuSiQue: its budgeted selection keeps the answer string less often (0.56 against 0.78; 0.31 against 0.44), and coverage falls by 0.29 and 0.15.

Controlled perturbations (pilot). Table 11 and Figure 2 (Appendix F) summarise Experiment 2. These comparisons are many (roughly fifty cells), and we expect a few intervals to exclude zero by chance; we report the patterns that recur across datasets or representations. When required facts are removed (P1), the model mostly abstains rather than guessing: abstention rises from 0.43 to 0.72 on MuSiQue (R1) and from 0.24 to 0.48 on 2WikiMultiHopQA. Hard distractors (P4) reduce exact match in every QA cell, by up to 0.16 (HotpotQA R1, $[-0.28, -0.04]$) and by 0.09 on MuSiQue R1 ($[-0.17, -0.01]$), and raise abstention on MuSiQue. Retyping or flipping 40% of the derived relations (P2, R4 only) lowers 2WikiMultiHopQA exact match by 0.10 $[-0.18, -0.01]$ on the 71 questions that have relations, and is inconclusive on MuSiQue and HotpotQA. Reordering (P3) and stripping source titles (P5) produce small effects of inconsistent sign, and HoVer and FEVER verdicts are almost unaffected by any intervention.

On the secondary datasets (50 questions each; run `C_exp0`), every Experiment 0 contrast is inconclusive, e.g. B4−B1 is 0.00 $[-0.12, +0.12]$ EM on HotpotQA and -0.06 $[-0.22, +0.12]$ verdict accuracy on HoVer.

F Additional results

Table 8 gives the absolute scores of every representation under Protocol A in the expanded pilot, Table 9 the full information accounting under Protocols A and B, Table 10 the retrieved-evidence baselines, and Table 11 and Figure 2 the perturbation experiment. Table 5 lists the secondary contrasts on the confirmatory sample and Table 7 the analysis of predicted contradiction lines; the remaining tables and figures are exploratory pilot results.

G Revision experiments: supplementary tables

Table 13 stratifies the contradiction-line effect by the number of such lines. Table 15 summarises the predicted contradictions. Table 16 gives the retrieved-evidence contrasts under both protocols, and Table 17 by retrieval outcome. Table 18 makes the budgets and costs of Experiment 0 explicit. All of these are exploratory. The blinded human-verification packages and their importers are described in the repository (`docs/REMAINING_HUMAN_VERIFICATION.md`).

Data	Contrast	n	ΔEM	95% CI	Ident.	ΔCov.
MuSiQue	R2−R1	1744	+0.013	[-0.001, +0.028]	2	+0.041
MuSiQue	R3−R5	1744	-0.007	[-0.016, +0.002]	0	-0.015
MuSiQue	R4P−R4	1744	-0.045	[-0.060, -0.030]	94	-0.089
2Wiki	R2−R1	3232	-0.024	[-0.036, -0.012]	4	-0.017
2Wiki	R3−R5	3232	-0.004	[-0.016, +0.008]	0	-0.015
2Wiki	R4P−R4	3232	-0.036	[-0.045, -0.027]	818	-0.061

Table 5: Secondary contrasts on the confirmatory sample (pre-specified as secondary; not part of the tested family, so descriptive). Source: `F_exp1_contrasts.csv`.

Data	Contrast	ΔEM	both ans.	one abstains	abst. only first / only second	neither
2Wiki	R1−R0	-0.017	+0.001	-0.018	250/81	529
2Wiki	R4−R5	+0.006	+0.010	-0.004	123/88	731
2Wiki	R4P−R4	-0.036	-0.011	-0.025	257/59	795
2Wiki	R5−R1	-0.008	-0.004	-0.004	170/130	649
MuSiQue	R1−R0	-0.001	-0.002	+0.001	94/110	710
MuSiQue	R4−R5	+0.010	+0.006	+0.004	82/89	709
MuSiQue	R4P−R4	-0.045	-0.005	-0.040	218/63	728
MuSiQue	R5−R1	-0.011	-0.012	+0.001	98/104	700

Table 6: Exploratory decomposition of each paired EM difference on the confirmatory sample (added after the confirmatory result). *both ans.*: contribution of questions answered under both representations; *one abstains*: contribution of questions answered under only one. The columns sum to ΔEM. Source: `F_exp1_decomposition_exploratory.csv`.

Data	R4P shows CONTRADICTS	n	Abst. R4	Abst. R4P	EM R4	EM R4P
2Wiki	yes	2034	0.32	0.41	0.33	0.28
2Wiki	no	1198	0.17	0.17	0.48	0.48
MuSiQue	yes	1010	0.50	0.63	0.25	0.18
MuSiQue	no	734	0.39	0.42	0.33	0.32

Table 7: Exploratory: the same questions under derived relations (R4) and predicted relations (R4P), split by whether the predicted block contains at least one `CONTRADICTS` line (an input property). Source: `F_exp1_r4p_relations_exploratory_paired.csv`.

Data	R0	R1	R2	R3	R4	R4P	R5
MuSiQue	0.28 (0.57)	0.31 (0.57)	0.28 (0.59)	0.24 (0.52)	0.29 (0.59)	0.21 (0.42)	0.29 (0.58)
2Wiki	0.44 (0.84)	0.37 (0.76)	0.36 (0.74)	0.37 (0.67)	0.41 (0.69)	0.38 (0.66)	0.38 (0.70)
HotpotQA	0.40 (0.88)	0.42 (0.82)	0.35 (0.85)	0.40 (0.86)	0.40 (0.84)	0.42 (0.80)	0.38 (0.82)
HoVer	0.68 (1.00)	0.68 (1.00)	0.73 (1.00)	–	0.68 (1.00)	0.74 (1.00)	0.68 (1.00)
FEVER	0.96 (1.00)	0.96 (1.00)	0.97 (1.00)	–	0.96 (1.00)	0.96 (1.00)	0.96 (1.00)

Table 8: Exploratory pilot, Protocol A: EM (QA) or verdict accuracy (HoVer, FEVER), with coverage (answered rate) in parentheses, for the same facts in seven representations. n=100 (MuSiQue, 2Wiki) or 50. R3 is unavailable without a reasoning order. Source: `C_exp1_absolute.csv`.

Data	Setting	Rep.	Kept	Ovh.	Tok.	Ans. drop	Ans. in ev.	EM	Cov.
MuSiQue	A	R0	1.00	33	378	0.00	0.99	0.28	0.57
MuSiQue	A	R1	1.00	122	467	0.00	0.99	0.31	0.57
MuSiQue	A	R5	1.00	122	467	0.00	0.99	0.29	0.58
MuSiQue	A	R4	1.00	141	486	0.00	0.99	0.29	0.59
MuSiQue	B 1.0× per-rep.	R0	0.81	32	327	0.19	0.89	0.25	0.52
MuSiQue	B 1.0× per-rep.	R1	0.65	81	322	0.28	0.85	0.26	0.52
MuSiQue	B 1.0× per-rep.	R5	0.65	81	322	0.28	0.85	0.22	0.53
MuSiQue	B 1.0× per-rep.	R4	0.62	88	321	0.35	0.80	0.17	0.44
MuSiQue	B 1.0× shared	R0	0.60	30	251	0.38	0.77	0.18	0.47
MuSiQue	B 1.0× shared	R1	0.60	75	296	0.38	0.77	0.20	0.48
MuSiQue	B 1.0× shared	R5	0.60	75	296	0.38	0.77	0.18	0.52
MuSiQue	B 1.0× shared	R4	0.60	86	307	0.38	0.77	0.17	0.46
MuSiQue	B 0.5× per-rep.	R0	0.34	23	144	0.71	0.49	0.10	0.30
MuSiQue	B 0.5× per-rep.	R1	0.30	38	143	0.77	0.43	0.09	0.30
MuSiQue	B 0.5× per-rep.	R5	0.30	38	143	0.77	0.43	0.09	0.30
MuSiQue	B 0.5× per-rep.	R4	0.29	39	142	0.79	0.42	0.08	0.31
MuSiQue	B 0.5× shared	R0	0.27	21	120	0.80	0.42	0.09	0.28
MuSiQue	B 0.5× shared	R1	0.27	35	133	0.80	0.42	0.09	0.30
MuSiQue	B 0.5× shared	R5	0.27	35	133	0.80	0.42	0.09	0.30
MuSiQue	B 0.5× shared	R4	0.27	38	136	0.80	0.42	0.08	0.30
2Wiki	A	R0	1.00	29	414	0.00	0.98	0.44	0.84
2Wiki	A	R1	1.00	157	543	0.00	0.98	0.37	0.76
2Wiki	A	R5	1.00	157	543	0.00	0.98	0.38	0.70
2Wiki	A	R4	1.00	167	552	0.00	0.98	0.41	0.69
2Wiki	B 1.0× per-rep.	R0	0.83	28	366	0.11	0.85	0.38	0.78
2Wiki	B 1.0× per-rep.	R1	0.64	104	361	0.33	0.66	0.33	0.67
2Wiki	B 1.0× per-rep.	R5	0.64	104	361	0.33	0.66	0.31	0.64
2Wiki	B 1.0× per-rep.	R4	0.63	108	361	0.34	0.64	0.33	0.64
2Wiki	B 1.0× shared	R0	0.60	26	265	0.39	0.60	0.30	0.67
2Wiki	B 1.0× shared	R1	0.60	97	337	0.39	0.60	0.27	0.63
2Wiki	B 1.0× shared	R5	0.60	97	337	0.39	0.60	0.27	0.62
2Wiki	B 1.0× shared	R4	0.60	103	342	0.39	0.60	0.29	0.62
2Wiki	B 0.5× per-rep.	R0	0.39	21	173	0.59	0.43	0.17	0.50
2Wiki	B 0.5× per-rep.	R1	0.32	52	171	0.62	0.39	0.17	0.39
2Wiki	B 0.5× per-rep.	R5	0.32	52	171	0.62	0.39	0.19	0.41
2Wiki	B 0.5× per-rep.	R4	0.31	52	170	0.62	0.39	0.15	0.38
2Wiki	B 0.5× shared	R0	0.29	20	131	0.63	0.38	0.14	0.45
2Wiki	B 0.5× shared	R1	0.29	48	160	0.63	0.38	0.15	0.36
2Wiki	B 0.5× shared	R5	0.29	48	160	0.63	0.38	0.16	0.37
2Wiki	B 0.5× shared	R4	0.29	50	161	0.63	0.38	0.15	0.37

Table 9: Information accounting under Protocol A and Protocol B (budget $\beta\times$ the content tokens of the fact set). Kept: fraction of facts shown; Ovh./Tok.: mean overhead and evidence tokens; Ans. drop: answer-bearing facts cut per question; Ans. in ev.: answer string present in the block. Source: `C_exp1_protocol_accounting.csv`, `C_exp1_absolute.csv`.

Data	Sys.	EM	95% CI	F1	Cov.	Tok.	Units	Ans. in ev.	Lat. (s)
MuSiQue	B0	0.06	[+0.02, +0.11]	0.10	0.23	493	4.0	0.34	0.36
MuSiQue	B1	0.12	[+0.06, +0.19]	0.17	0.33	538	4.0	0.44	0.36
MuSiQue	B2	0.13	[+0.07, +0.20]	0.17	0.36	1390	57.1	0.44	0.86
MuSiQue	B3	0.15	[+0.08, +0.22]	0.18	0.38	1667	57.1	0.44	0.91
MuSiQue	B4	0.03	[+0.00, +0.07]	0.05	0.18	442	11.9	0.31	0.51
2Wiki	B0	0.23	[+0.16, +0.31]	0.27	0.54	506	4.0	0.69	0.38
2Wiki	B1	0.26	[+0.18, +0.35]	0.30	0.54	501	4.0	0.78	0.39
2Wiki	B2	0.24	[+0.16, +0.33]	0.27	0.48	1436	56.4	0.78	0.67
2Wiki	B3	0.20	[+0.12, +0.28]	0.24	0.50	1711	56.4	0.78	0.68
2Wiki	B4	0.11	[+0.05, +0.17]	0.13	0.25	489	11.9	0.56	0.51

Table 10: Experiment 0, retrieved evidence (exploratory, n=100 per dataset). Tok.: mean evidence tokens; Units: items shown; Lat.: answer-generation latency (claim extraction for B2–B4 is not included). B4 is our reconstruction of EG-RAG. Source: `A_exp0_absolute.csv`.

Data	Rep.	Perturbation	n	Lab.-pres.	ΔEM	95% CI	Ans. changed	Abstention
MuSiQue	R1	P1	100	0	–	–	0.40	0.43→0.72
MuSiQue	R1	P3	100	100	-0.07	[-0.13, -0.02]	0.13	0.43→0.44
MuSiQue	R1	P4 hard	100	100	-0.09	[-0.17, -0.01]	0.27	0.43→0.51
MuSiQue	R1	P5 strip	100	100	-0.03	[-0.08, +0.02]	0.12	0.43→0.41
MuSiQue	R1	P6 entity	100	0	–	–	0.14	0.43→0.50
MuSiQue	R1	P6 negation	29	0	–	–	0.03	0.52→0.55
MuSiQue	R4	P1	100	0	–	–	0.44	0.41→0.71
MuSiQue	R4	P2	100	100	-0.01	[-0.07, +0.05]	0.24	0.41→0.47
MuSiQue	R4	P3	100	100	-0.03	[-0.08, +0.02]	0.14	0.41→0.44
MuSiQue	R4	P4 hard	100	100	-0.08	[-0.16, +0.00]	0.34	0.41→0.54
MuSiQue	R4	P5 strip	100	100	+0.00	[-0.04, +0.04]	0.10	0.41→0.38
MuSiQue	R4	P6 entity	100	0	–	–	0.19	0.41→0.43
MuSiQue	R4	P6 negation	29	0	–	–	0.03	0.48→0.52
2Wiki	R1	P1	100	0	–	–	0.43	0.24→0.48
2Wiki	R1	P3	100	100	+0.04	[-0.04, +0.12]	0.30	0.24→0.19
2Wiki	R1	P4 hard	100	100	-0.04	[-0.13, +0.05]	0.35	0.24→0.31
2Wiki	R1	P5 strip	100	100	+0.02	[-0.03, +0.08]	0.16	0.24→0.18
2Wiki	R1	P6 entity	100	0	–	–	0.14	0.24→0.27
2Wiki	R1	P6 negation	15	0	–	–	0.00	0.20→0.20
2Wiki	R4	P1	100	0	–	–	0.43	0.31→0.51
2Wiki	R4	P2	71	71	-0.10	[-0.18, -0.01]	0.44	0.35→0.34
2Wiki	R4	P3	100	100	-0.01	[-0.09, +0.07]	0.26	0.31→0.26
2Wiki	R4	P4 hard	100	100	-0.03	[-0.12, +0.06]	0.35	0.31→0.36
2Wiki	R4	P5 strip	100	100	+0.08	[+0.01, +0.16]	0.19	0.31→0.26
2Wiki	R4	P6 entity	100	0	–	–	0.17	0.31→0.33
2Wiki	R4	P6 negation	15	0	–	–	0.00	0.40→0.40
HotpotQA	R1	P1	50	0	–	–	0.36	0.18→0.44
HotpotQA	R1	P3	50	50	-0.06	[-0.16, +0.04]	0.18	0.18→0.14
HotpotQA	R1	P4 hard	50	50	-0.16	[-0.28, -0.04]	0.26	0.18→0.24
HotpotQA	R1	P5 strip	50	50	-0.02	[-0.08, +0.04]	0.08	0.18→0.14
HotpotQA	R1	P6 entity	50	0	–	–	0.18	0.18→0.18
HotpotQA	R1	P6 negation	5	0	–	–	0.00	0.20→0.20
HotpotQA	R4	P1	50	0	–	–	0.42	0.16→0.44
HotpotQA	R4	P2	25	25	-0.04	[-0.24, +0.12]	0.28	0.08→0.16
HotpotQA	R4	P3	50	50	-0.04	[-0.14, +0.06]	0.14	0.16→0.16
HotpotQA	R4	P4 hard	50	50	-0.10	[-0.24, +0.04]	0.34	0.16→0.20
HotpotQA	R4	P5 strip	50	50	+0.02	[-0.04, +0.08]	0.08	0.16→0.14
HotpotQA	R4	P6 entity	50	0	–	–	0.16	0.16→0.18
HotpotQA	R4	P6 negation	5	0	–	–	0.00	0.20→0.20

Table 11: Experiment 2 at rate 0.4 (exploratory). ΔEM is computed only on label-preserving records without introduced answers (Lab.-pres.); P1 and P6 can remove the answer, so only answer-change and abstention rates are given. Source: `C_exp2_sensitivity.csv`.

Data	Change (second vs. first)	n	ΔEM	ΔAbst.	ΔWrong	ΔAcc. | both answered	C→A / A→C	C→W / W→C	W→A / A→W
2Wiki	sentence items vs. passages	3232	-0.017	+0.052	-0.036	+0.002 [-0.013, +0.017]	87/29	152/156	163/52
2Wiki	graph order vs. dataset order	3232	-0.008	+0.012	-0.004	-0.006 [-0.018, +0.005]	57/45	95/81	113/85
2Wiki	derived relation lines vs. none	3232	+0.006	+0.011	-0.017	+0.014 [+0.005, +0.024]	42/29	42/74	81/59
2Wiki	R4P vs. R4 (original)	3232	-0.036	+0.061	-0.025	-0.017 [-0.027, -0.007]	97/17	80/44	160/42
2Wiki	contradiction lines on vs. off	3232	-0.033	+0.061	-0.028	-0.013 [-0.022, -0.004]	89/9	64/36	137/20
MuSiQue	sentence items vs. passages	1744	-0.001	-0.009	+0.010	-0.004 [-0.020, +0.014]	42/44	29/26	52/66
MuSiQue	graph order vs. dataset order	1744	-0.011	-0.003	+0.015	-0.025 [-0.045, -0.006]	45/46	43/22	53/58
MuSiQue	derived relation lines vs. none	1744	+0.010	-0.004	-0.006	+0.013 [-0.007, +0.031]	28/35	29/40	54/54
MuSiQue	R4P vs. R4 (original)	1744	-0.045	+0.089	-0.044	-0.012 [-0.037, +0.011]	93/24	43/34	125/39
MuSiQue	contradiction lines on vs. off	1744	-0.033	+0.062	-0.029	-0.009 [-0.020, +0.003]	52/2	14/7	64/6

Table 12: Question-level outcome transitions (C correct, W answered and wrong, A not answered) between paired conditions on the confirmatory questions (exploratory decomposition). ΔAcc. | both answered: accuracy difference among questions answered under both conditions, with 95% CI. Sources: `revision/analysis/F_exp1_transitions.csv`, `revision/controls/controls_transitions.csv`.

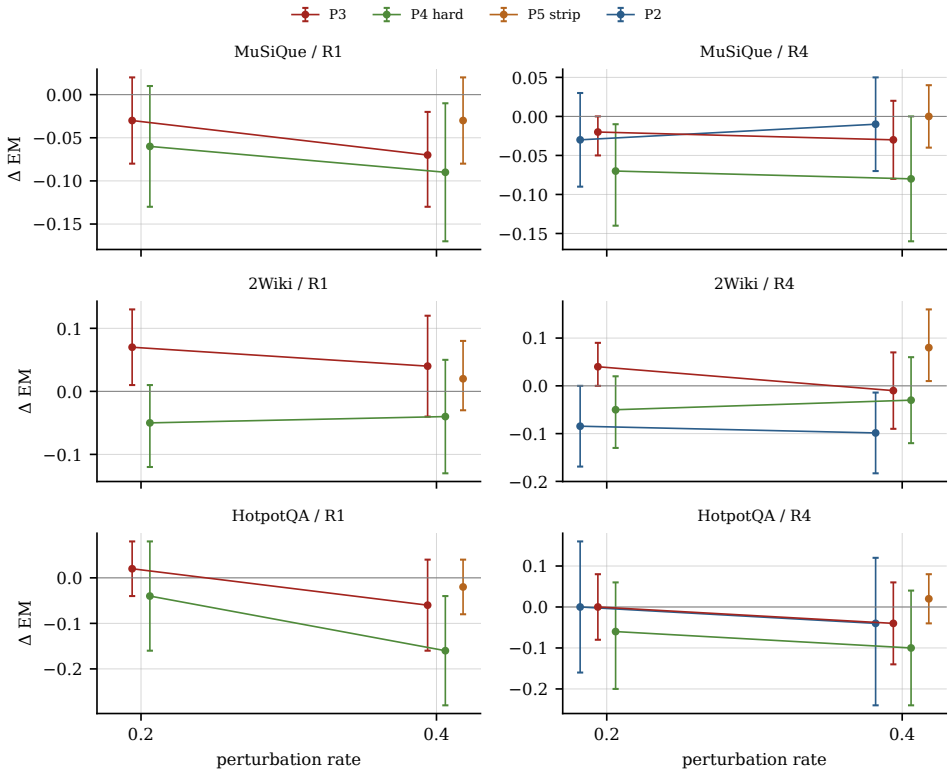


Figure 2: Exploratory pilot: change in exact match under label-preserving perturbations, relative to the same question unperturbed (95% CIs). P1 and P6 can remove the answer and are reported in Table 11 only.

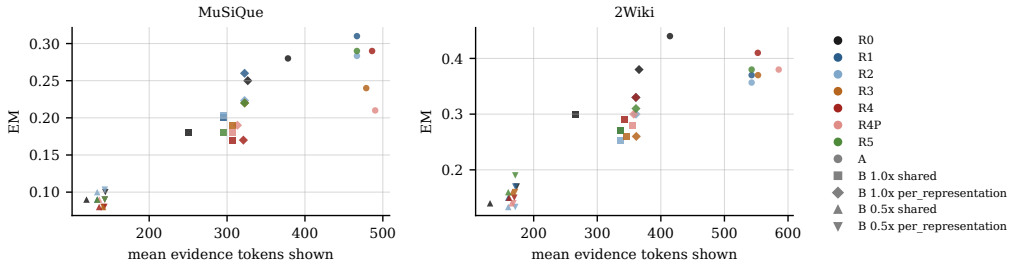


Figure 3: Exploratory pilot: exact match against the mean number of evidence tokens shown, for Protocol A (natural length) and Protocol B (budget $1.0\times$ or $0.5\times$ the content tokens; shared or per-representation policy).

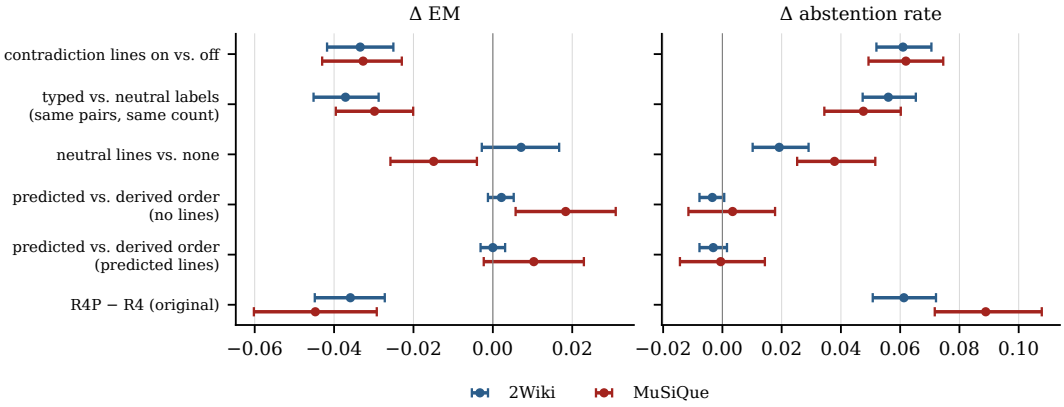


Figure 4: Fixed-order relation controls on the confirmatory questions (exploratory). Each row changes one factor on identical facts; points are paired differences with 95% CIs.

Data	Contradiction lines	n	ΔEM	95% CI	ΔAbst.	95% CI
2Wiki	0	1198	+0.000	[+0.000, +0.000]	+0.000	[+0.000, +0.000]
2Wiki	1-2	1015	-0.069	[-0.089, -0.049]	+0.088	[+0.068, +0.108]
2Wiki	3-5	577	-0.050	[-0.075, -0.024]	+0.137	[+0.106, +0.166]
2Wiki	6+	442	-0.020	[-0.043, +0.002]	+0.066	[+0.034, +0.097]
MuSiQue	0	734	+0.000	[+0.000, +0.000]	+0.000	[+0.000, +0.000]
MuSiQue	1-2	660	-0.053	[-0.073, -0.033]	+0.094	[+0.071, +0.118]
MuSiQue	3-5	257	-0.039	[-0.066, -0.016]	+0.132	[+0.089, +0.179]
MuSiQue	6+	93	-0.129	[-0.194, -0.065]	+0.129	[+0.065, +0.204]

Table 13: Effect of showing the predicted `CONTRADICTS` lines (R4P_FO−R4P_FO_NC) by the number of such lines in the prompt (exploratory). With no contradiction lines the two prompts are identical. Source: `revision/controls/relation_control_strata.csv`.

Prompt	Contrast	n	ΔEM	95% CI	ΔAbst.
`qa-v2`	R1−R0	1000	-0.023	[-0.044, -0.002]	+0.059
`qa-v2`	R4P−R4	1000	-0.036	[-0.054, -0.020]	+0.054
`noabst`	R1−R0	1000	-0.004	[-0.026, +0.018]	+0.002
`noabst`	R4P−R4	1000	-0.017	[-0.035, +0.000]	+0.003
prompt effect	`noabst`−`qa-v2` (R0)	1000	+0.078	[+0.055, +0.101]	-0.196
prompt effect	`noabst`−`qa-v2` (R1)	1000	+0.097	[+0.072, +0.123]	-0.253
prompt effect	`noabst`−`qa-v2` (R4)	1000	+0.109	[+0.084, +0.135]	-0.268
prompt effect	`noabst`−`qa-v2` (R4P)	1000	+0.128	[+0.103, +0.153]	-0.319
interaction	(R1−R0) `noabst` minus `qa-v2`	1000	+0.019	[-0.006, +0.045]	-0.057
interaction	(R4P−R4) `noabst` minus `qa-v2`	1000	+0.019	[-0.001, +0.040]	-0.051

Table 14: Prompt factor (exploratory, 1,000 confirmatory 2Wiki questions): the frozen prompt `qa-v2` and the same prompt without the line offering *insufficient evidence* (`noabst`); last rows give the prompt × representation interaction. Source: `revision/noabst/noabst_contrasts.csv`.

Automatic triage (heuristic, not a judgement)	Pairs	Share
different source titles (different entities)	3,857	40%
same title, different years	1,962	20%
same title, different numbers	671	7%
same title, pronoun-initial sentence	734	8%
same title, other	2,526	26%
all predicted contradiction pairs	9,750	100%

Table 15: Predicted `CONTRADICTS` pairs on the confirmatory sample (R4P; 12,816 directed edges, 12,372 rendered, 3,044 questions). NLI confidence ≥0.90 (median 0.985). Categories are heuristic triage for prioritising human verification. Source: `revision/contradictions/contradictions_summary.json`.

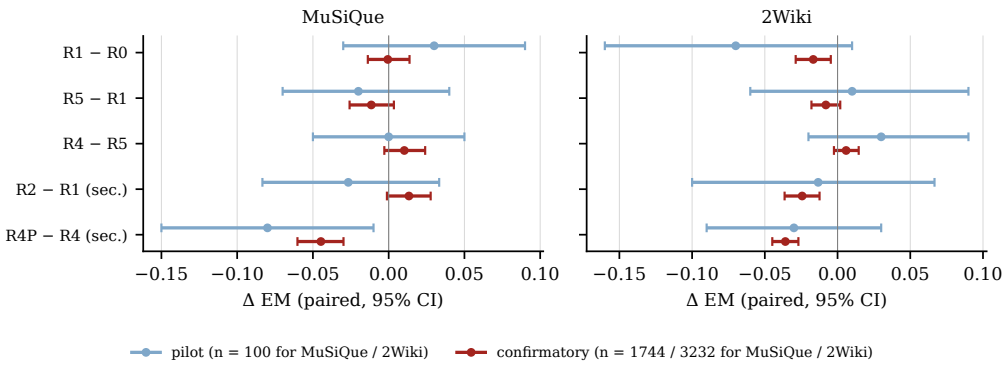


Figure 5: Protocol A contrasts in the exploratory pilot (n=100 per dataset) and on the disjoint confirmatory sample. The pilot estimate of R1−R0 on 2WikiMultiHopQA (−0.07) overstated the confirmed effect (−0.017) by a factor of four.

Data	Prot.	Contrast	n	Ident.	ΔEM	95% CI	ΔAbst.	Tokens
MuSiQue	A	R1−R0	500	0	+0.014	[+0.000, +0.028]	-0.006	526→653
MuSiQue	A	R5−R1	500	451	+0.010	[+0.000, +0.020]	-0.008	653→653
MuSiQue	A	R4−R5	500	342	+0.002	[-0.012, +0.016]	-0.002	653→657
MuSiQue	A	R4P−R4	500	72	-0.036	[-0.056, -0.020]	+0.068	657→701
MuSiQue	A	contradiction lines on−off	500	131	-0.022	[-0.038, -0.008]	+0.040	665→701
MuSiQue	A	R4P_FO−R5	500	86	-0.022	[-0.038, -0.008]	+0.052	653→701
MuSiQue	B 1.0×	R1−R0	500	3	+0.022	[+0.002, +0.042]	+0.014	462→457
MuSiQue	B 1.0×	R5−R1	500	459	+0.002	[-0.004, +0.008]	-0.002	457→457
MuSiQue	B 1.0×	R4−R5	500	370	+0.006	[-0.004, +0.018]	-0.006	457→457
MuSiQue	B 1.0×	R4P−R4	500	133	-0.032	[-0.050, -0.016]	+0.052	457→452
MuSiQue	B 1.0×	contradiction lines on−off	500	201	-0.016	[-0.028, -0.006]	+0.026	455→452
MuSiQue	B 1.0×	R4P_FO−R5	500	144	-0.020	[-0.038, -0.004]	+0.038	457→452
2Wiki	A	R1−R0	500	5	-0.026	[-0.054, +0.002]	+0.072	525→681
2Wiki	A	R5−R1	500	498	+0.000	[+0.000, +0.000]	+0.000	681→681
2Wiki	A	R4−R5	500	322	+0.010	[-0.008, +0.028]	-0.006	681→685
2Wiki	A	R4P−R4	500	84	-0.030	[-0.054, -0.008]	+0.066	685→749
2Wiki	A	contradiction lines on−off	500	110	-0.024	[-0.046, -0.004]	+0.068	695→749
2Wiki	A	R4P_FO−R5	500	66	-0.016	[-0.042, +0.010]	+0.052	681→749
2Wiki	B 1.0×	R1−R0	500	23	+0.010	[-0.016, +0.034]	+0.030	454→451
2Wiki	B 1.0×	R5−R1	500	499	+0.000	[+0.000, +0.000]	+0.000	451→451
2Wiki	B 1.0×	R4−R5	500	383	-0.002	[-0.014, +0.010]	+0.004	451→451
2Wiki	B 1.0×	R4P−R4	500	157	-0.040	[-0.060, -0.020]	+0.054	451→447
2Wiki	B 1.0×	contradiction lines on−off	500	195	-0.036	[-0.054, -0.018]	+0.044	451→447
2Wiki	B 1.0×	R4P_FO−R5	500	141	-0.044	[-0.066, -0.022]	+0.064	451→447

Table 16: Retrieved evidence (exploratory, fresh disjoint sample of 500 questions per dataset): every representation renders the same four reranked passages; R4/R5 use annotation-derived edges restricted to retrieved sentence pairs (oracle structure on real retrieval). Protocol B: budget $1.0\times$ content tokens, per-representation policy. Source: `revision/retrieval/retrieval_contrasts.csv`.

Data	Contrast	Stratum	n	ΔEM	95% CI	ΔAbst.
MuSiQue	R1−R0	retrieval: complete	149	+0.034	[-0.007, +0.074]	-0.034
MuSiQue	R1−R0	retrieval: partial	311	+0.006	[-0.006, +0.019]	+0.010
MuSiQue	R1−R0	retrieval: none	40	+0.000	[+0.000, +0.000]	-0.025
MuSiQue	R1−R0	distractor heavy: True	328	+0.006	[+0.000, +0.015]	+0.012
MuSiQue	R1−R0	long evidence: True	249	+0.012	[-0.008, +0.032]	-0.020
MuSiQue	contradiction lines on−off	retrieval: complete	149	-0.047	[-0.094, -0.007]	+0.047
MuSiQue	contradiction lines on−off	retrieval: partial	311	-0.013	[-0.026, -0.003]	+0.039
MuSiQue	contradiction lines on−off	retrieval: none	40	+0.000	[+0.000, +0.000]	+0.025
MuSiQue	contradiction lines on−off	distractor heavy: True	328	-0.012	[-0.024, -0.003]	+0.034
MuSiQue	contradiction lines on−off	long evidence: True	249	-0.016	[-0.040, +0.004]	+0.036
2Wiki	R1−R0	retrieval: complete	262	-0.057	[-0.099, -0.015]	+0.080
2Wiki	R1−R0	retrieval: partial	238	+0.008	[-0.025, +0.042]	+0.063
2Wiki	R1−R0	distractor heavy: True	119	+0.008	[+0.000, +0.025]	+0.067
2Wiki	R1−R0	long evidence: True	250	-0.036	[-0.072, +0.000]	+0.060
2Wiki	contradiction lines on−off	retrieval: complete	262	-0.019	[-0.050, +0.011]	+0.069
2Wiki	contradiction lines on−off	retrieval: partial	238	-0.029	[-0.055, -0.008]	+0.067
2Wiki	contradiction lines on−off	distractor heavy: True	119	-0.008	[-0.025, +0.000]	+0.050
2Wiki	contradiction lines on−off	long evidence: True	250	-0.012	[-0.040, +0.016]	+0.072

Table 17: Retrieved evidence, Protocol A, by retrospective retrieval outcome (gold labels used only to stratify). Exploratory; small strata have wide intervals. Source: `revision/retrieval/retrieval_contrasts.csv`.

Data	Sys.	EM	Cov.	Tok.	Units	Gold-pass. rec.	Ans. in ev.	Extr. calls	Latency (s)	CONTRA share
MuSiQue	B0	0.06	0.23	493	4.0	0.44	0.34	0.0	0.4	–
MuSiQue	B1	0.12	0.33	538	4.0	0.53	0.44	0.0	0.4	–
MuSiQue	B2	0.13	0.36	1390	57.1	0.53	0.44	15.3	24.3	–
MuSiQue	B3	0.15	0.38	1667	57.1	0.53	0.44	15.3	24.4	–
MuSiQue	B4	0.03	0.18	442	11.9	0.55	0.31	15.1	23.8	0.55
2Wiki	B0	0.23	0.54	506	4.0	0.76	0.69	0.0	0.4	–
2Wiki	B1	0.26	0.54	501	4.0	0.80	0.78	0.0	0.4	–
2Wiki	B2	0.24	0.48	1436	56.4	0.80	0.78	14.7	23.9	–
2Wiki	B3	0.20	0.50	1711	56.4	0.80	0.78	14.7	23.9	–
2Wiki	B4	0.11	0.25	489	11.9	0.80	0.56	14.7	23.8	0.62

Table 18: Experiment 0 with budgets and costs made explicit (pilot, n=100 per dataset). Budgets are *not* matched: B2/B3 show about three times the tokens of B0/B1; B4 selects at most 12 claims within 450 tokens. Latency includes LLM claim extraction (original-call latency for cached calls). CONTRA share: fraction of B4's predicted relations labelled `CONTRADICTS`. Source: `revision/analysis/C_exp0_costs.csv`.